S-BLOCK

Density ⇒ Li < Na > K < Rb < Cs

SRP ⇒ -3.04 (Li) ↓ -2.7 (Na) ↓ -2.9 (K, Rb)

MP/BP Li > Na > K > Rb > Cs

IE/H.E/Radius Li < Na < K < Rb < Cs

Cs & K → electrodes in photoelectric cell

$Li + O_2 \rightarrow Li_2O$ oxide

$Na + O_2 \rightarrow Na_2O_2 + Na_2O$ per.

$K/Rb/Cs + O_2 \rightarrow MO_2 + M_2O_2$ super.

only $Li + N_2 \rightarrow Li_3N$

Although SRP Li > Na

Na > Li vigorous with H_2O

$M^+ + \bar{O}H + H_2$ ←

$M + (x+y)NH_3 \rightarrow [M(NH_3)_n]^+ + [e(NH_3)_y]^-$

blue paramag. ammoniated e⁻

* on standing, → $MNH_2 + H_2\uparrow$

* conc^t soln ⇒ blue → bronze diamag.

Li → white metal use

KOH → absorb CO_2

$M_2O + H_2O \rightarrow MOH$

$M_2O_2 + H_2O \rightarrow MOH + H_2O_2$

$MO_2 + H_2O \rightarrow MOH + H_2O_2 + O_2$

oxides/peroxides ⇒ colourless

superoxides ⇒ yellow/orange.

Alk-earth in liq amm → deep blue black ←

ΔH_f° ⇒ LiCl < NaCl < — (I or Br) —

more -ve

LiF > NaF > ----

more -ve

MF > MCl > ----

more -ve

MP/BP ⇒ MF > MCl > ---

LiCl is soluble in pyridine and ethanol

$Li_2CO_3 \xrightarrow{\Delta} Li_2O + CO_2$

other $M_2CO_3 \xrightarrow{\Delta}$ xxx

$LiHCO_3(s)$ does not exist

$LiCl \cdot 2H_2O$ exists like $MgCl_2 \cdot 8H_2O$

Li is much harder than others

Na↓ — HC≡CH — ↓Li

$NaC{\equiv}CNa$ xxxx

$LiNO_3 \xrightarrow{\Delta} Li_2O + NO_2 + O_2$

$MNO_3 \xrightarrow{\Delta} MNO_2 + O_2$

Size $Mg^{2+} < Li^+$

$Mg > Li$

Flame colour

Li → Crimson red

Na → Yellow

K → Violet

Rb → Red violet

Cs → Blue

Be → x

Al → x

Ca → brick red

Sr → Crimson

Ba → Apple green

Best way to prep BeF_2

$(NH_4)_2BeF_4 \xrightarrow{\Delta}$

and $BeCl_2 + CO$ ← $BeO + C + Cl_2$

$BeCl_2 + LiAlH_4 \rightarrow BeH_2$

others react directly w/ H_2.

BeO ⇒ covalent amphoteric

others ⇒ ionic & basic

Solid: Be, Cl, Cl, Be, Cl, Cl, Be — chain

Vapour (same for $AlCl_3$): Be, Cl, Cl, Be, Cl, Cl — dimer

> 1200K linear Cl–Be–Cl

hydrated halides of Be/Mg undergo hydrolysis on Δ.

Ca/Sr/Ba halides → dehydration

$BeCO_3$ unstable. Kept only in atm of CO_2.

$MgCl_2 \cdot 8H_2O$ $CaCl_2 \cdot 6H_2O$

$SrCl_2 \cdot 6H_2O$ $BaCl_2 \cdot 2H_2O$

$Mg(NO_3)_2 \cdot 6H_2O$ $Ba(NO_3)_2$

$M(NO_3)_2 \xrightarrow{\Delta} MO + NO_2 + O_2$

(like $LiNO_3$) brown gas

Be + acids → oxide film

↳ alkali. $[Be(OH)_4]^{2-}$ beryllate

Bio: ⊙ Na⇔K pump in cell membrane consumes $\frac{1}{3}$ ATP ≈ 15 kg in 24 hours.

⊙ Activate enzymes

⊙ Mg as cofactor.

⊙ Chlorophyll ⇒ Mg

⊙ Calcitonin, parathyroid regulate Ca.

Cement: CaO + SiO_2 + Clay + oxides of Al, Fe, Mg

silica (SiO_2)

Portland ⇒ CaO 50-60%; SiO_2 20%.

Al_2O_3 5-10%, MgO, Fe_2O_3, SO_3 1-2% each

ratio: $\frac{SiO_2}{Al_2O_3} \approx$ 2.5 to 4 $\frac{CaO}{\text{oxides}} \approx 2$

COMPOUNDS OF CALCIUM

Compound	Formula	Formed By	Formed in	Properties	Reacⁿ involved	Uses
Calcium Oxide	CaO	$CaCO_3 \xrightarrow{\Delta} CaO + O_2$	Rotatory Kiln	Absorbs Moisture $CaO + CO_2 \rightarrow CaCO_3$	$CaO + H_2O \rightarrow Ca(OH)_2$ slaking of lime $CaO + SiO_2 \rightarrow CaSiO_3$ $6CaO + P_4H_{10} \rightarrow 2Ca_3(PO_4)_2$ Sodium Sodalime $CaO + 3C \rightarrow CaC_2 + CO\uparrow$ Calcium Carbide	Manufacturing of Cement
Calcium Carbonate	$CaCO_3$ (Limestone)	$Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$ $CaCl_2 + Na_2CO_3 \rightarrow CaCO_3 + 2NaCl$	Excess must be avoided दोहरा → $(CaHCO_3)_2$ soluble Ban Jayega	CO_2 is Evolved Used as an Antacid.	$CaCO_3 \xrightarrow{\Delta} CaO + CO_2$ With Acid $CaCO_3 + HCl \rightarrow CaCl_2 + H_2O + CO_2$ $CaCO_3 + H_2SO_4 \rightarrow CaSO_4 + CO_2 + H_2O$	Manufacture of Quick lime $CaCO_3 + MgCO_3 \rightarrow$ Flux
POP	$CaSO_4 \cdot 1/2 H_2O$	Gypsum $CaSO_4 \cdot 2H_2O$ ↓ Formed by heating Further heating → 393K dead burnt plaster ($CaSO_4$)		Hemihydrate of Calcium sulphate. Setting property on mixing with high water	(fracture Banane ke Kaam Aata hai)	Building industry Immobilising the affected part of Organ
Cement		$CaO + SiO_2 +$ Clay + oxides ↓ silica [Al, Fe, Mg] Raw materials: Limestone, Clay	Limestone + Clay ↓ Heat ↓ Cement Klinker mixed with 2-3% weight of Gypsum ↓ Cement	Called Portland Cement Gypsum is added to slow down setting property → Reason Hydration of Molecules with constituent particles of H_2O	Ingredients: Dicalcium Silicate + Tricalcium Silicate (Ca_3SiO_5) Ca_2SiO_4 Tricalcium Aluminate ($Ca_3Al_2O_6$)	Construction in Dams, Bridges.
Slaked lime (Calcium Hydroxide)	$Ca(OH)_2$	$CaO + H_2O \rightarrow Ca(OH)_2$		Aqueous sol^n → Lime Water Suspension in: Milk of water lime Disinfectant in Nature	$2Ca(OH)_2 + 2Cl_2 \rightarrow CaCl_2 + Ca(OCl)_2 + 2H_2O$ Milk of lime → Hypochlorite → constituent of Bleaching Powder $Ca(OH)_2 + CO_2 \rightarrow CaCO_3$ Milky When Exs is passed $CaCO_3 + CO_2 + H_2O \rightarrow Ca(HCO_3)_2$ ↓ soluble	White wash due to disinfectant nature.

AYUSHMAN BISWART

Compound	Formula & Name	Formed by	Properties	Uses
1) Sodium Carbonate (Na_2CO_3)	$Na_2CO_3 \cdot 10H_2O$ (Washing Soda) Decahydrate	Solvay Process. Ammonium is recovered. $2NH_3 + H_2O + CO_2 \rightarrow (NH_4)_2CO_3$ $(NH_4)_2CO_3 + H_2O + CO_2 \rightarrow 2NH_4HCO_3$ $NH_4HCO_3 + NaCl \rightarrow NH_4Cl + NaHCO_3$ $2NaHCO_3 \xrightarrow{\Delta} Na_2CO_3 + CO_2 + H_2O$	K_2CO_3 can't be formed ↓ $KHCO_3$ is >> soluble than KCl ↓ hence during precipitation solubility of $NaHCO_3$ << NaCl $2NH_4Cl + Ca(OH)_2 \rightarrow 2NH_3 + CaCl_2 + H_2O$ ↓ Recovering NH_3	Laboratory Reagent, Water softening $Na_2CO_3 \cdot 10H_2O \xrightarrow{\Delta} Na_2CO_3 \cdot H_2O + 9H_2O$ $Na_2CO_3 \cdot 10H_2O \xrightarrow{\Delta \text{ further}} Na_2CO_3 + H_2O$ Soda Ash
2) Sodium chloride	NaCl, common salt	Impurities: Ca, Mg chlorides, Ca/Mg sulphates. Crystallization of Brine soln.	Hygroscopic. Reacn with $K_2Cr_2O_7$ + conc. H_2SO_4 $4NaCl + K_2Cr_2O_7 + 5H_2SO_4 \xrightarrow{\Delta} 4NaHSO_4 + K_2SO_4 + 2CrO_2Cl_2 + 3H_2O$ ↓ orange red vapour	Common salt, manufacture NaOH, Na_2CO_3
3) Sodium Hydroxide $NaOH \xrightarrow{\text{alkali metal}}$ no reacn	NaOH (Caustic Soda)	Electrolysis of NaCl in Castner Kellner cell → Brine soln Electrolysed.	Deliquescent (having tendency to become liquid) Na Amalgum + water → $NaOH + 2Hg + H_2\uparrow$ $NaOH + CO_2 \rightarrow Na_2CO_3$	Laboratory Reagent
4) Sodium Bicarbonate	$NaHCO_3$ (Baking Soda)	$Na_2CO_3 + H_2O + CO_2 \rightarrow 2NaHCO_3$	White crystalline. Decomposes on heating, producing bubbles of CO_2	Skin infection, Fire Extinguisher. Cakes for making Pt fluffy. $NaHCO_3 \rightarrow Na_2CO_3 + CO_2 + H_2O$

Mortar: slaked lime (1 Part) and Sand (3 Part)

COMPOUNDS OF SODIUM

AYUSHMAN BISWARI